CO2 - AI - 2:-

1923111 77

G. L. Mahesh Kumar

1. Regular Expression Design for input validation:-

1. Binary sting that End with 0

R.E = (0+1)* 01

2. Binary sting containing substing 11 or 00

R.E = (0+1)* (11+00) (0+1)*

Explanation:- Accepts binary sting containing Eaithr 11 (or) 00 anyware in the sting.

3. sting over containing atleast one accerance of ab

R.E = (a+b)* ab(a+b)*

4. sting over 1 having an Even number of zeros

R.E = 1*(01*01)*

5. stings over that stant with a and End with b.

R.E = a(a+b)* b

2. Regular Expression analysis and language Representation:

Given regular expression R = (0+1)* 101(0+1)*

This regular Expresson accepts all binary stings that conting the substing 101 at least once.

1. Identify the alphabet and substrings:-

Alphabet Σ = 1

Recognated substring : 101

2. Five accepted strings :-

101, 0101, 1010, 11010, 00101.

3. three rejected strings :-

0001, 0010, 11100,

4. Kleen star (*)

Represents zero (or) more representation of symbols.

Union (+)

Represents a choice of zeros and or 1's.

3. Content free Grammar construction :-

Arthametic Expression involving identifiers and additions.

start Symbol : E

Terminate symbol : {a, b, +}

non-terminate symbol : {E}

Production rules : E → E+E / a / b.

Balanced Parenthesis :-

language :- strings containing properly balanced opening and closing parenthesis.

start Symbol : S

Terminate Symbols : {c, }}

non-Terminate Symbols : {S}

production rules :- S → (S)S / S

Sol:- ε, (), (()), ()(), (()())

$S \rightarrow 0S1S/1S0S/\varepsilon$

Result:- the content - free grammers were sucessfully Constructed.

Derivation and parse tree construction:-

Given grammar: $S \rightarrow aSb/\varepsilon$

(1) components of Grammar.

Start symbol : S

Terminal symbols: {a, b}

non - Terminal symbols: {S}

(2) # Generate string:-

n	string
0	ε
1	ab
2	aabb
3	aaabbb

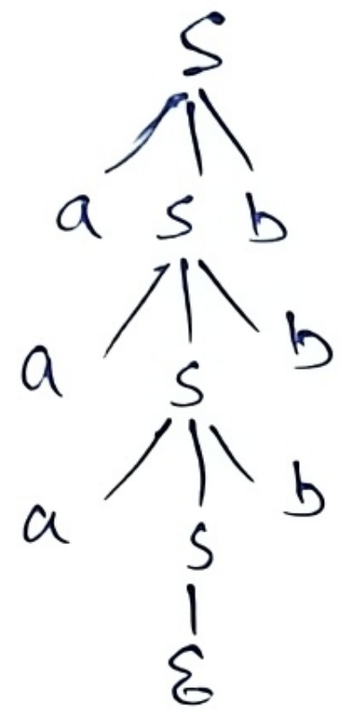
(3) left most derivative of aaabbb:

$S \Rightarrow asb$
$\Rightarrow aasb$
$\Rightarrow aaasbbb$
$\Rightarrow aaabbb$

4. Right most Derivate of aaabbb:

$S \Rightarrow asb$
$\rightarrow aasb$
$\rightarrow aaasbbb$

5. Parse tree.



5. Grammar Design for Arithmetic Expressions:-

(1) productions rules:- $E \rightarrow E+T \mid T$

$T \rightarrow T*F \mid F$

$F \rightarrow (E) \mid id$

(2) Components of Grammer:

Start symbol: S

Terminate symbol: id, +, *, (,)

non-terminate Symbol: {E, T, F}

4. Derivation of a + b*c:

$E \rightarrow E+T$

$\Rightarrow T+T$

$\Rightarrow F+T$

$\Rightarrow id+T$

$\Rightarrow id+T*F$

$\Rightarrow id+F*F$

$\Rightarrow id+id*F$

46/50

5. Parse tree:-

